Which Side Is Wrong? Neutral Contradiction Flags Do Not Help Retrieval-Augmented Generation Against Misinformation

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Abstract

Retrieval-augmented generation systems increasingly detect contradictions among retrieved passages and flag them to the generator. We ask what such a flag should say, and what the detector must get right for it to help. In a pre-registered study on RAMDocs, we vary the detector (heuristic, NLI, LLM, random, oracle) and the format (a neutral flag, a directional flag naming the likely misinformation, or removal of the suspect passage), and add synthetic detectors of controlled precision, recall and direction accuracy. Even a perfectly accurate neutral flag does not help: it lowers gpt-4o-mini's accuracy on conflict questions by 6.7 points, and its estimate is negative for all three generators we test (gpt-4o-mini, gpt-4.1, Qwen2.5-7B) under two prompts, whereas the same flag helps on the genuine conflicts of WikiContradict. Acting on direction helps: removing the passage the oracle names gains 10.8–30.2 points in every setting, and naming it in the prompt beats a neutral flag everywhere, though it beats no flag only for generators that follow the warning. Removal is fragile: false flags delete correct passages, and a wrong direction does as much harm as a neutral flag or more. With a real LLM detector, a directional flag beats a neutral one by 16.5 points on conflict questions, but its net gain over no flags is small. We release the harness, the pre-registration and every run.

1 Introduction

Retrieval-augmented generation (RAG) (Lewis et al., 2020) hands a language model passages that may disagree. Some disagreements are genuine: two sources report different findings, or two people share a name. Others are planted or stale misinformation (Zou et al., 2025; Wang et al., 2025b). A natural fix is to detect the contradiction and tell the generator about it. Recent systems pair pairwise contradiction detectors with prompts that flag conflicting passages (Wang et al., 2026), and prompting a model about conflicts can help (Cattan et al., 2025). These systems report detection F1 and end-to-end accuracy, but not how the two connect, which hides two questions. What should the flag say? And what must the detector get right for the flag to help?

We separate the two. We fix the retrieved passages and vary, independently, the *detector* (a heuristic, an NLI cross-encoder, an LLM, a random placebo and an oracle) and the *format* that passes its output to the generator: a neutral "these passages may conflict" flag, a *directional* flag that names the likely misinformation, or removal of the suspect passage. To see what detector quality buys, we also build synthetic detectors with a chosen precision, recall and direction accuracy. We ran pilots on 200 RAMDocs questions (Wang et al., 2025b), then pre-registered our hypotheses and tested them on 300 held-out questions with three generators and two prompt skeletons, plus a contrast benchmark of genuine conflicts (Hou et al., 2024).

Our findings:

- **A correct neutral flag does not help against misinformation.** Flagging exactly the true contradictions lowers gpt-4o-mini's accuracy on RAMDocs conflict questions by 6.7 points $[-10.8, -3.1]$. The estimate is negative for all three generators and both prompt skeletons, and significant in four of the six settings. The same flag *raises* accuracy on WikiContradict, where both sides are valid, by 6.5 points (19.0 with gpt-4.1).

- **Acting on direction helps; telling the model helps if it listens.** Removing the passage the oracle names as misinformation gains 10.8–30.2 points on conflict questions in every setting. A directional flag beats the neutral flag everywhere, but beats no flag only for generators that follow the warning (gpt-4.1; gpt-4o-mini with the default prompt). Naming the wrong side

costs 7.0 points as a flag and 21.8 with removal.

- **Removal is the high-variance choice.** False flags make the model lose correct passages. With a real LLM detector, removal's 18.7-point gain on conflict questions is offset by a 20.6-point loss on clean ones, in every generator.
- **Real detectors gain little so far** (exploratory). With the same LLM detector, a directional flag beats a neutral one by 16.5 points on conflict questions, but its net effect over no flags is +4.9 points on labelled questions. The heuristic detector we started from does not beat random flags of the same volume, and a grid of synthetic detectors predicts real detectors' effects only roughly.

For conflict-aware RAG, we recommend reporting direction accuracy and effects on conflict-free questions alongside detection F1, under the exact format that feeds the generator. We release the harness, pre-registration and every run.[1]

2 Related Work

Knowledge conflicts. Language models can give priority to memorised knowledge over contradicting context (Longpre et al., 2021). They are also readily persuaded by coherent external evidence that contradicts their memory (Xie et al., 2024), and they adopt wrong context over a correct prior to a degree that depends on how strong that prior is (Wu et al., 2024). Xu et al. (2024) distinguish context–memory, inter-context and intra-memory conflicts. We study inter-context conflicts: disagreements among retrieved passages, as in Chen et al. (2022), who find that models tend to rely on a single passage when the evidence conflicts.

Benchmarks with conflicting evidence. RAMDocs (Wang et al., 2025b) mixes genuine ambiguity (several valid answers), misinformation and noise across retrieved documents, and builds on AmbigDocs (Lee et al., 2024). CONFLICTS (Cattan et al., 2025) annotates search results with a taxonomy of conflict types, each with its own desired behaviour. WikiContradict (Hou et al., 2024) pairs Wikipedia passages that genuinely contradict each other. FaithEval (Ming et al., 2025), ConflictBank (Su et al., 2024) and ConfRAG (Yuan et al., 2026) cover related settings. We use RAMDocs because its document labels say which side of a contradiction is the misinformation, and WikiContradict as a contrast case where neither side is.

Telling the model about conflicts. The closest prior work tells the generator that a conflict exists. Cattan et al. (2025) find that prompting a model to reason about the conflict type helps, and that supplying the gold conflict type helps much more. On WikiContradict, models acknowledge both sides more often when told the passages may conflict (Hou et al., 2024). Ragability (Gross et al., 2026) shows the cost of such a hint: it improves conflict identification on contradictory contexts but hurts it on *consistent* ones, where the hint is false. Our neutral-flag result is different: the hint is true, the input is contradictory, and accuracy still falls, because the hint does not say which side to trust. Pan et al. (2023b) test misinformation detection and "vigilant" prompting as defences against LLM-generated misinformation in open-domain QA, and Pan et al. (2023a) call for QA that is aware of misinformation. Source signals also matter: models weigh retrieved pages' metadata and appearance (Chiang and Lee, 2024), and passing source-credibility signals to a retrieval-augmented fact-checker helps resolve conflicting evidence (Ge et al., 2025). We separate the *content* of the hint (neutral, directional, removal) from its *accuracy* (precision, recall, direction), and compare neutral and directional hints head-to-head with the same detector.

Conflict detection, resolution and poisoning. ConflictRAG (Wang et al., 2026) trains a pairwise detector with an LLM fallback and reports pair-level F1 alongside answer gains; its NLI and chain-of-thought baselines resemble two of our detectors. LLMs are imperfect context validators (Gokul et al., 2025). TCR (Ye et al., 2026) passes conflict signals through a learned soft prompt, EvidentialRAG (Hossain et al., 2026) routes between answering and abstaining, and credibility-aware generation (Pan et al., 2024) fine-tunes models to weigh per-passage credibility, a trained analogue of our directional flags. MADAM-RAG (Wang et al., 2025b), Astute RAG (Wang et al., 2025a), Self-RAG (Asai et al., 2024) and Corrective RAG (Yan et al., 2024) handle unreliable passages without an explicit contradiction detector. A few injected passages can steer a RAG system (Zou et al., 2025); RobustRAG (Xiang et al., 2024) isolates passages for certifiable robustness, and TrustRAG

[1] Code, the pre-registration and all run outputs are in the anonymised supplementary material and will be released publicly on acceptance.

(Zhou et al., 2025) filters suspected poisoned ones, the analogue of our removal format. Our question is orthogonal: given a detector's output, which way of passing it to the generator helps, and what must the detector get right?

3 Framework

A question q comes with retrieved passages $P = (p_0, \ldots, p_{n-1})$. A *detector* returns flagged pairs $F \subseteq \{(i, j) : i < j\}$. A *directional* detector also returns, for each flagged pair, a *suspect* set $S_{ij} \subseteq \{i, j\}$: the side it believes is misinformation. A *format* turns (P, F, S) into the generator's prompt. We vary detectors and formats independently. Every prompt shares a skeleton: passages, question, and an instruction to end with a `Final answer(s):` line. The *default* skeleton asks for every valid answer when the passages support several (RAMDocs contains same-name ambiguity, so a correct answer may need several entities). The *minimal* skeleton drops that policy, and its answer line allows several answers without inviting them (Appendix A).

3.1 Formats

No flags (plain). Passages and question only.

Instruction. Plain plus a generic instruction to check the passages for contradictions, with no detector. We test two wordings. v1 ends "if a disagreement reflects genuinely different valid answers, report each one"; v2 asks the model to judge which passages are reliable and to leave out answers supported only by unreliable ones.

Flag. Each flagged passage gets the header `[POTENTIAL CONFLICT with passage(s): j]`, followed by a list `Passages i and j: possible contradiction`. This is the format of the original CAAG system, except that every detector's flags are rendered identically (no detector-specific type labels or confidence scores), so detectors differ only in *which* pairs they flag.

Directional flag. As Flag, but suspect passages are headed `[LIKELY MISINFORMATION; conflicts with passage(s): j]`, and the list says which passage likely contains misinformation.

Remove (suspect). The suspect passages of every flagged pair are deleted from the prompt. For a pair without a suspect, the second passage is deleted, which is what the original CAAG REMOVE strategy does when there are no retrieval scores.

3.2 Detectors

CAAG (Contradiction-Aware Attention Gating) is a heuristic detector from an earlier, unpublished RAG system, used here with its code unchanged. A pair is scored only if its MiniLM (Reimers and Gurevych, 2019) cosine similarity reaches $\theta_s = 0.65$. Its score is the sum of a negation and contrast asymmetry term, a named-entity mismatch term (differing dates, numbers, money or percentages) and a temporal-marker term, multiplied by the similarity. A pair is flagged if the score reaches $\theta_c = 0.35$. It runs locally and gives no direction.

NLI is `nli-deberta-v3-base` (He et al., 2023), a cross-encoder scored in both directions; a pair is flagged if the larger contradiction probability exceeds 0.5. It gives no direction.

LLM is a single gpt-4o-mini call (temperature 0) that lists contradicting pairs, telling them apart from ambiguity, complementary and temporal relations, and names the side that is more likely misinformation (or "both" or "unclear"), as JSON.

Random flags each pair independently at CAAG's per-pair flag rate on the confirmatory split (6.22%). It is a placebo that matches CAAG's flag volume.

Oracle flags exactly the gold contradiction pairs and names the gold misinformation side. The *inverted oracle*, an adversarial control, flags the same pairs but names the correct side.

3.3 Synthetic detectors

To separate a detector's accuracy from its identity, we build detectors with a chosen pair-level precision p, recall r and direction accuracy d from the gold labels. Each gold contradiction pair is flagged with probability r. False flags are drawn from non-contradicting pairs at a single per-pair rate, set over the whole split so that expected precision is p. That rate is dataset-wide, so conflict-free questions receive false flags too. A true flag names the gold misinformation side with probability d, and otherwise the other side. A false flag has no misinformation side, so its suspect is picked uniformly. Each pair gets one uniform draw for flagging and one for direction, and every cell reuses them. Cells are therefore nested: raising r or d,

or lowering p, only adds flags or corrections. This keeps comparisons across cells paired.

3.4 Placing real detectors on the grid

Can the grid predict a real detector's effect from a few summary numbers? We compare two ways of summarising a detector. *Pair-level*: its pair precision, recall and direction accuracy. *Document-level*: what it does to each subset's documents in a given format. For conflict questions, that is the share whose misinformation documents are all marked and the share that have a correct document marked; for clean questions, the share that have a correct document marked. Here "marked" means removed (Remove) or headed as misinformation (Directional flag). The prediction for a subset is the inverse-distance-weighted mean effect of the four nearest grid cells in that space.

4 Experimental Setup

Data. **RAMDocs** (Wang et al., 2025b) has 500 questions. Each has 1–12 documents (5.5 on average) typed *correct* (supports a gold answer, possibly for a different same-name entity), *misinfo* (supports a wrong answer) or *noise*. We label a pair a *contradiction* if one side is misinformation and the two answers differ, *ambiguity* if both are correct with different answers, *consistent* if the answers match, and *unrelated* if either is noise. A question is a *conflict* question if it has at least one contradiction pair, and *clean* otherwise.

The released types are imperfect. 163 documents typed *correct* contain a wrong answer; some are injected misinformation ("Ronald Powell, a renowned chess player"), others mention it incidentally (a navigation menu listing "Soccer"). We therefore leave pairs that touch these documents out of pair-level metrics. Questions whose only sign of conflict is such a document (55 of all 500; 35 of the confirmatory split) belong to neither subset; they count only in "All". Treating every such document as misinformation instead is a robustness check.

WikiContradict (Hou et al., 2024) has 253 questions, each with two Wikipedia passages that genuinely contradict each other. The reference answer is *both* sides and no passage is misinformation, so it is the contrast case to RAMDocs. An answer is scored correct if both of the dataset's answer strings (one per passage) match a final answer, with the bidirectional matching rule below. **CONFLICTS** (Cattan et al., 2025): we use the 237 questions that have a reference answer (outdated-information, misinformation and no-conflict items), which can be scored automatically.

Pre-registration and splits. We developed the method on a random 200-question pilot split of RAMDocs. Those pilots motivated directional flags, removal and document-level placement. Before running anything on the remaining 300 questions (139 conflict, 126 clean, 35 excluded), we committed a pre-registration that fixes the conditions, metrics, hypotheses and corrections. Every RAMDocs result in the main text is on this confirmatory split. Pilot results appear only in the appendix and are marked exploratory. Comparisons that were not registered, including every real-detector row of Table 1, are exploratory. After a review of a first draft we registered an addendum (H10, H11 and four re-analyses), before running anything it covers; Appendix C lists every deviation.

Generators. The main generator is `gpt-4o-mini` (OpenAI et al., 2024) (temperature 0.1, at most 256 output tokens, seeds 0–2). We replicate the key conditions with `gpt-4.1` (3 seeds) and with the open-weights Qwen2.5-7B-Instruct (Qwen Team et al., 2024) (4-bit, run locally with Ollama, seed 0), keeping the LLM detector fixed at gpt-4o-mini. Every prompt uses the *default* skeleton, which asks for every valid answer when the passages support several. To test whether our findings depend on that policy, we rerun the oracle and LLM-detector conditions with a *minimal* skeleton that drops it (Appendix A). MADAM-RAG (Wang et al., 2025b) is ported unchanged to the same client: three debate rounds, temperature 0, 128 tokens per call.

Metrics. On RAMDocs an answer is correct if it includes every gold answer and no wrong answer (Wang et al., 2025b). Our primary metric reads "includes" literally: after normalisation (lowercasing; removing punctuation and articles), each gold or wrong answer must appear as a whole-word span of the extracted final answer. A more lenient bidirectional variant, which also counts a predicted answer contained in a gold one, is our robustness metric. With the one-way metric, our port of MADAM-RAG scores 28.5% on the pilot split and 32.8% on all 500 questions (published:

28.00, on an undisclosed subset); with the bidirectional metric it scores 38.4% on all 500. We chose the one-way metric as primary for this reason, before the confirmatory run; the first draft of the pre-registration had named the bidirectional one. Because the published subset and scorer are unknown, we do not claim to reproduce the published number, and we compare conditions only within our own harness. On WikiContradict an answer is correct if it contains both sides; on CONFLICTS, if it contains the reference answer.

Statistics. We average each question's score over the three seeds, then report paired differences from the plain format with cluster-bootstrap 95% CIs (10,000 resamples) and two-sided sign-flip permutation tests (10,000 permutations) (Efron and Tibshirani, 1993; Dror et al., 2018). Holm correction (Holm, 1979) is applied within each pre-registered hypothesis family. A hypothesis is *robustly supported* if it also holds under the robustness metric. A null difference is not read as equivalence unless a two one-sided test (TOST; Lakens, 2017) with a pre-set margin confirms it. With 139 conflict questions, the smallest effect our comparisons detect with 80% power is 4–12 points (median 5.8; Card et al., 2020); smaller differences may go unseen.

Cost. Token counts come from the API's usage reports. Prices were checked on 2026-10-27 (gpt-4o-mini: \$0.15, \$0.075 cached, and \$0.60 per million input, cached-input and output tokens; gpt-4.1: \$2.00, \$0.50 and \$8.00). All experiments in this paper, pilots included, cost \$33.70 in API fees; Qwen ran locally.

5 Results

Table 1 gives every condition on the RAMDocs confirmatory split, Table 3 gives the outcome of every pre-registered test, and Figure 1 plots each condition's change on conflict questions against its change on clean ones.

5.1 A correct neutral flag does not help against misinformation

Flagging exactly the gold contradictions, in the neutral format, *lowers* gpt-4o-mini's accuracy on conflict questions by 6.7 points [−10.8, −3.1], from 22.1% (H2). With gpt-4.1 the drop is 8.6 points (H9). Both hold under either metric. The errors are not refusals. With the flag, the share of conflict answers that include the misinformation answer rises from 47.7% to 61.9%, while gold recall barely moves (72.6% to 69.8%): the model lists the misinformation *alongside* the correct answer.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	29.9	31.3	22.1	41.5	911
Prompt only					
Instruction (v1)	+1.6	+0.8	−1.2	+2.9	948
Instruction (v2)	+3.1	+2.3	+5.3	−1.1	951
Oracle detector					
Oracle, flag	**−3.1**	−3.5	**−6.7**	0.0	980
Oracle, directional flag	**+6.2**	**+7.0**	**+13.4**	0.0	998
Oracle, remove suspect	**+12.8**	**+14.5**	**+27.6**	0.0	825
Real detectors (exploratory)					
Random, flag	+1.0	+1.0	+0.7	+1.3	949
CAAG, flag	−0.4	−0.8	−0.7	−0.8	942
CAAG, remove	−3.3	−3.8	0.0	−7.9	809
NLI, flag	+0.2	+0.3	−0.5	+1.1	1021
NLI, remove	−5.7	−5.0	−2.2	−8.2	592
LLM, flag	−1.8	−1.3	−4.1	+1.9	970
LLM, directional flag	+5.0	+4.9	+12.5	−3.4	983
LLM, remove suspect	+0.3	0.0	**+18.7**	**−20.6**	718
Reference					
MADAM-RAG	+5.7	+5.7	+3.6	+7.9	8196

Table 1: RAMDocs hold-out, gpt-4o-mini, 3 seeds, one-way strict accuracy (primary). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

Not an artefact of the prompt. Our default skeleton asks for every valid answer, which could invite exactly this behaviour. Under the minimal skeleton, which drops that policy, the oracle neutral flag still lowers accuracy on conflict questions in all three generators (Table 2): by 5.3 points with gpt-4o-mini and 11.0 with gpt-4.1 (both significant after Holm; H10), and by 5.8 with Qwen2.5-7B (not significant). With the default skeleton Qwen's drop is also 5.8 points (not significant; H11). The estimate is negative in all six generator–skeleton settings. What changes is the mechanism. Without the answer policy, gpt-4o-mini no longer adds the misinformation (33.3% to 32.4%) but drops the correct answer (gold recall 57.1% to 47.1%). A neutral flag says that something is wrong without saying what, and the model either keeps both sides or loses the right one.

The same flag helps on genuine conflicts. On WikiContradict, where the reference answer is

		Plain	Oracle, conflict q.			LLM, directional flag		LLM, remove suspect	
Generator	Skeleton	conflict	flag	dir. flag	remove	conflict	clean	conflict	clean
gpt-4o-mini	default	22.1	**−6.7**	**+13.4**	**+27.6**	**+12.5**	−3.4	**+18.7**	**−20.6**
gpt-4o-mini	minimal	20.4	−5.3	+5.0	**+18.9**	+3.6	**−10.3**	+11.0	**−19.6**
gpt-4.1	default	21.8	**−8.6**	**+18.2**	**+30.2**	+8.4	0.0	**+20.6**	**−24.3**
gpt-4.1	minimal	33.6	**−11.0**	**+12.7**	**+14.9**	–	–	–	–
Qwen2.5-7B	default	19.4	−5.8	+3.6	**+18.0**	+2.9	−7.1	+7.9	**−18.3**
Qwen2.5-7B	minimal	17.3	−5.8	+2.9	**+10.8**	–	–	–	–

Table 2: The main effects across three generators and two prompt skeletons (RAMDocs hold-out, 3 seeds, one-way strict accuracy). *Plain*: accuracy (%) on conflict questions with no flags. Other columns: paired difference from plain in points. *Default* skeleton tells the model to give every valid answer; *minimal* drops that policy (Appendix A). **Bold**: significant after Holm across the table. The LLM detector is gpt-4o-mini for every generator. Qwen2.5-7B-Instruct runs locally (4-bit).

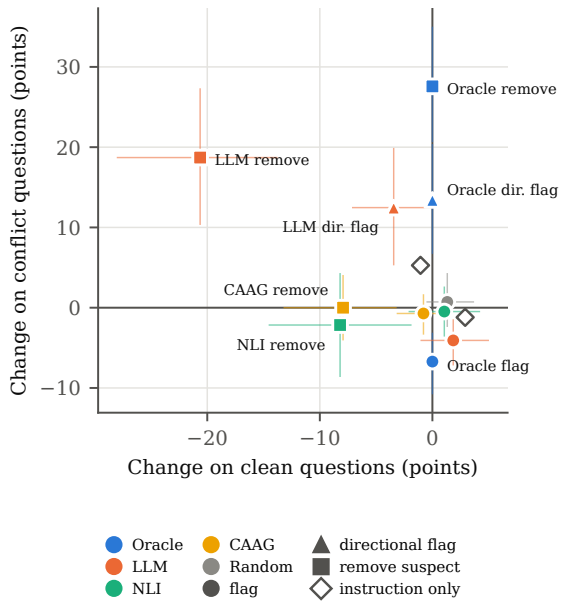


Figure 1: Change in one-way strict accuracy on RAMDocs conflict questions (y) against clean questions (x), relative to no flags (gpt-4o-mini, default skeleton, confirmatory split, 95% CIs). Colour: detector; shape: format. The oracle sits on the y-axis because it never flags a clean question. Without direction, removal deletes the second passage of each pair.

both sides, the oracle flag *raises* the share of answers that give both sides by 6.5 points [+3.0, +10.0] with gpt-4o-mini, and by 19.0 points [+14.5, +23.8] with gpt-4.1 (H8; Table 10). A neutral flag passes on *that* a conflict exists but not *which kind* it is, and whether it helps depends on whether keeping both sides is the right behaviour.

5.2 Acting on direction helps; a directional flag helps when the model follows it

Given the same oracle pairs, naming the misinformation side beats the neutral flag on conflict questions in all six generator–skeleton settings (H3, H9–H11; Table 3). Against no flags, *removing* the suspect passage gains in every setting, from 10.8 to 30.2 points, all significant after Holm. The directional *flag* is less consistent. It gains 13.4 points with gpt-4o-mini [+6.7, +20.4] and 18.2 with gpt-4.1 under the default skeleton, and 12.7 with gpt-4.1 under the minimal one. But with gpt-4o-mini under the minimal skeleton (+5.0) and with Qwen2.5-7B (+3.6, +2.9) the gain is not significant. Qwen largely ignores the header: the share of its answers that include the misinformation falls only from 48.2% to 42.4%, against 47.7% to 14.9% for gpt-4o-mini.

Where the directional flag fails (exploratory). After seeing the minimal-skeleton run, we split conflict questions by the number of gold answers. 105 of 139 have several (same-name ambiguity *and* misinformation), and a strict answer must list all of them. On the 34 single-answer conflict questions, the directional flag's estimate is positive in every setting, from +11.8 points (Qwen) to +46.1 (gpt-4o-mini), both with the default skeleton. On multi-answer questions it changes little except with gpt-4.1 (gpt-4o-mini +2.9, and −5.7 with the minimal skeleton; gpt-4.1 +19.0). One reading is that a warning about one passage makes the smaller models more conservative about the legitimate alternative answers too; we did not test this. Removal does not depend on the model acting on a warning, and it gains in both groups.

A wrong direction (exploratory). An inverted oracle that names the *correct* side lowers gpt-4o-mini's accuracy on conflict questions by 7.0 points as a directional flag [−11.5, −2.9], about as much as a neutral flag, and by 21.8 [−28.8, −15.3] when the correct passage is removed. Direction is a

strong lever in both directions, which makes it an attack surface (Ethics).

5.3 What a detector must get right

Figure 2 replaces the oracle with synthetic detectors of known quality. Two thresholds appear. First, *direction accuracy*: at a direction accuracy of 0.5 or 0.75, no cell gains significantly on all questions in either format. Only perfect direction does (directional flag $+6$, removal $+13$ points at perfect precision). Second, *precision*, but mainly for removal. On clean questions a cell's prompt does not depend on direction accuracy, so the false-flag cells reduce to 9 distinct comparisons per format. Removal is significantly harmful in 6 of them, the directional flag in 0 (H4a): false flags make the model delete correct passages, which costs clean questions 15 points at precision 0.5 and 26 at 0.25, while synthetic false directional flags cost at most two points. The harm tracks the share of clean questions that lose a correct document (Spearman ρ =-0.72 over 24 distinct cells, 95% CI [-0.86, -0.19] from a question-level bootstrap; H4b). Removal hurt clean questions significantly more than the directional flag in 5 of 9 matched cells (H4c; we set no threshold, so we report the count). Removal is the more powerful and the more fragile interface: its best cell gains twice the directional flag's on all questions, but below precision 0.75 it is the only format that does net harm.

5.4 Real detectors (exploratory)

Table 4 (Appendix B) describes the detectors. The pre-registered F1 ordering (H1) holds only in part. NLI beats CAAG (23.8 vs. 11.6), but the LLM detector's F1 (26.4) is not significantly higher than NLI's. Its direction accuracy is 88.3% (95% CI [83.3, 92.9], clustered by question). NLI and the LLM detector flag most clean questions (61.9% and 76.2%); CAAG flags 34.1%.

In the neutral format no real detector helps. CAAG's flags do not beat random flags at the same rate (H5; -0.4 vs. $+1.0$ points overall). The 90% CI of the difference, [-3.1, $+0.1$], rules out a gain larger than $+0.1$ points but does not establish equivalence within ± 3 points (TOST).

Direction helps with a real detector too. Holding the LLM detector's pairs fixed, the directional flag beats the neutral flag on conflict questions by 16.5 points [+9.6, +23.7] ($p < 0.001$; registered as exploratory in the addendum). Against no flags, the directional flag gains 12.5 points on conflict questions (raw $p = 0.002$, not significant after Holm across Table 1) and costs 3.4 on clean ones (6.1 under the bidirectional metric). Over the 265 labelled questions its net effect is $+4.9$ [$+0.8$, $+9.3$]. Removal gains 18.7 on conflict questions and loses 20.6 on clean ones, netting 0.0 [-6.2, $+6.0$]. With gpt-4.1 as the generator, the directional flag's gain over all 300 questions ($+5.4$) is significant after Holm, but the 35 unlabelled questions contribute disproportionately to it (they gain 13.3 points); over the labelled questions it is $+4.4$ [$+0.6$, $+8.3$]. Real false flags cost clean questions more than synthetic ones, probably because they concentrate on correct–correct pairs (Section 6). For reference, MADAM-RAG gains 5.7 points overall at roughly nine times the prompt tokens. CAAG in its original typed and scored format behaves the same as in the uniform format (Appendix B).

5.5 The grid predicts real detectors only roughly

If the grid only described synthetic detectors, it would be of limited use. Figure 3 (Appendix B) tests whether it predicts real ones (H6). As pre-registered, we bootstrap the whole prediction (cell effects, every detector's location and the interpolation) over questions, and ask whether the actual effect falls inside the 95% interval. Locating detectors by what they do to documents (the share of conflict questions whose misinformation is marked, and the share of questions that have a correct document marked) puts 7 of 8 effects inside; locating them by pair precision, recall and direction accuracy puts 6 inside. Document-level errors are smaller on average (4.1 vs. 5.3 points), but the paired difference, -1.2 [-3.6, $+0.4$] (exact sign-flip $p = 0.41$), is not significant over 8 cases. Both spaces miss the same case, the LLM detector's removal gain on conflict questions: one correctly directed pair is enough to remove a misinformation document, so the detector removes the misinformation on 61.2% of conflict questions despite its low pair recall. We chose the document-level axes after seeing the LLM detector on the pilot split. The grid is a useful first estimate of a detector's effect, but with three detectors we cannot claim that one summary predicts better than the other.

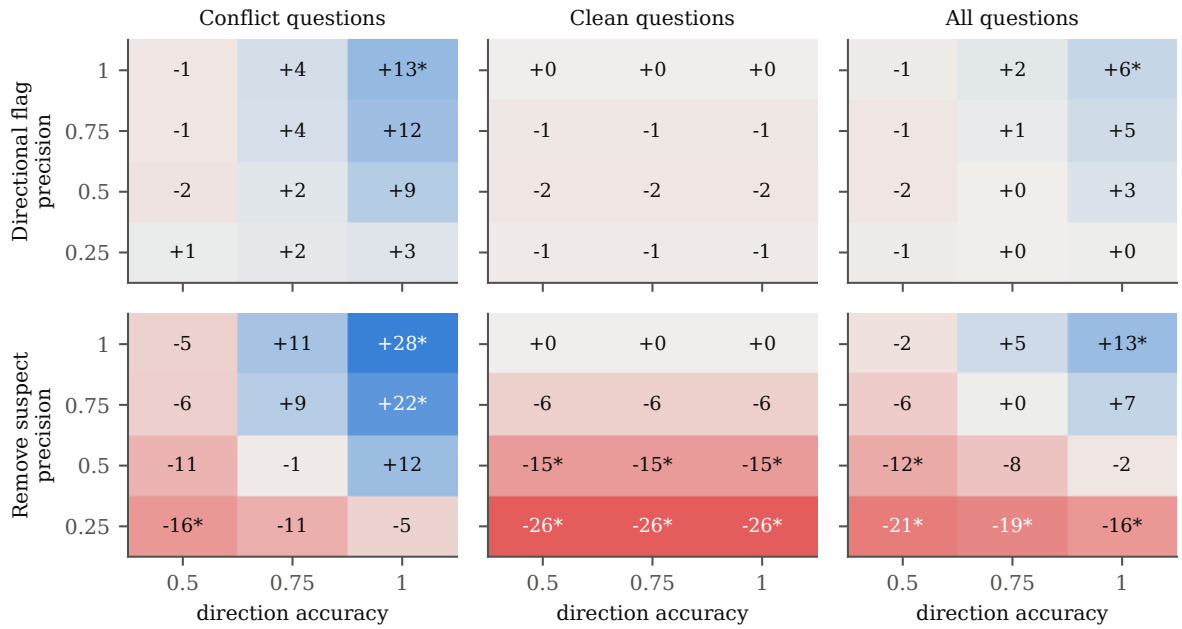


Figure 2: Synthetic detectors at recall 1: change in one-way strict accuracy (points) against no flags, by pair precision (rows) and direction accuracy (columns), for the directional flag (top) and removal (bottom). *: significant after Holm correction over the 36 cells of that mode and subset. Clean questions only see false flags, so their rows do not vary with direction accuracy.

5.6 Prompt-only baselines and CONFLICTS

Neither instruction wording changes RAMDocs accuracy significantly after correction (H7). On WikiContradict, v1 (which allows "genuinely different valid answers") raises both-sides answers by 8.7 points and v2 (which asks the model to drop unreliable passages) lowers them by 6.7: each wording builds in one policy, right for one kind of conflict and wrong for the other. On the 237 automatically scorable CONFLICTS questions (exploratory, one seed; most have no conflict), no condition moves accuracy significantly (Appendix B).

6 Discussion

Why neutral flags do not help. A neutral flag says that the passages disagree but not which one to trust, so the model falls back on its answer policy: with the default prompt it keeps both sides, without it it more often drops the correct one. Either way a strict answer to a misinformation question loses, while on WikiContradict, where both sides are right, the same flag helps. This sharpens the finding of Cattan et al. (2025) that knowing the conflict *type* helps: for misinformation, the useful part of the type is the direction. It also extends Ragability's observation that false hints hurt (Gross et al., 2026) to true hints that are not specific enough.

When direction helps. Removal helps with every generator because it needs nothing from the model; a directional flag helps only if the model acts on the warning, and a wrong direction is at least as harmful as a neutral flag. Practitioners should check that a directional flag changes their generator's behaviour before relying on it.

Implications for detectors and interfaces. Detection F1 alone does not say what a detector will do downstream: with the same LLM detector, adding direction raised conflict-question accuracy by 16.5 points over a neutral flag, and real false flags, concentrated on correct–correct pairs, cost clean questions more than synthetic ones. We recommend reporting (i) direction accuracy, (ii) how often correct documents are marked or removed on conflict-free questions, and (iii) effects on conflict-free questions, under the exact format used downstream. Neither interface is safe with today's detectors. With our LLM detector, removal gains 18.7 points on conflict questions and loses 20.6 on clean ones; the directional flag still costs clean questions 3.4 (default skeleton) to 10.3 points (minimal). Removal should be gated on a measured clean-question loss rate.

Limitations

Models. We test three generators: two closed OpenAI models and one 7B open-weights model, run at 4-bit precision with one seed. Larger open models and reasoning models may respond differently, especially to the directional header. The LLM detector is a single prompt to one model. We compare against no trained detector such as ConflictRAG's or TrustRAG's filter (Wang et al., 2026; Zhou et al., 2025), which would sit at different points on the grid.

Data and power. The confirmatory evidence comes from one benchmark, RAMDocs: 300 questions, 139 of them conflict questions. Its misinformation is synthetic, and its document labels are imperfect. We exclude 35 hold-out questions whose status depends on documents we could not label without human annotation, and we check that the conclusions survive treating those documents as misinformation. Our comparisons detect effects of about 5.8 points with 80% power, so smaller effects, including small harms on clean questions, may be missed. WikiContradict has genuine conflicts but no misinformation side. CONFLICTS, the most realistic benchmark we use, shows no significant effect for any condition. It was run with one seed, scored only on items with a reference answer, and without the planned LLM judge, so it can neither confirm nor refute the RAMDocs findings. We do not know how the findings transfer to naturally occurring misinformation.

Metric. RAMDocs ships no scorer. Our primary metric reconstructs the published definition and scores our MADAM-RAG port close to the published number on the pilot split, but the published subset is unknown, so this is not a reproduction. The metric is string matching: it misses paraphrases, and partial answers interact with "no wrong answer". We report a more lenient variant throughout and mark any conclusion that depends on the choice.

Direction and the grid. Oracle direction is close to giving the model an answer key, so the oracle rows are upper bounds. In the synthetic grid only perfect direction gives a significant overall gain. The real LLM detector, at 88.3% direction accuracy, gains on conflict questions over a neutral flag, but its net gain over no flags is significant only with gpt-4.1. Synthetic false flags are spread uniformly and their suspect side is random, whereas real detectors concentrate their errors. The grid predicts real detectors only roughly, and three detectors cannot show whether document- or pair-level summaries predict better.

Scope. We study flags passed to the generator as text. Learned interfaces such as soft prompts or credibility fine-tuning (Ye et al., 2026; Pan et al., 2024), and retrieval-side filtering, may follow different trade-offs. We test one adversarial case, a detector that always names the wrong side, but not an attacker who controls which pairs are flagged.

Ethics Statement

This work uses public benchmarks and commercial model APIs, and involves no human subjects or personal data. Its main risk is dual use. Our results show that a false "likely misinformation" label, or the removal of a passage, can change a RAG system's answers: an inverted oracle that names the correct side as misinformation costs up to 21.8 points. An adversary who can trigger a deployed detector could exploit this (Zou et al., 2025). We therefore report harm on conflict-free questions alongside every gain, and we argue that systems should measure what a detector does at the document level before acting on its flags.

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A Prompts

All prompts below are copied verbatim from the code that produced the results.

Generation skeleton (default).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Minimal skeleton (no answer policy).

```
Answer the question using the passages below.
If no passage answers the question, the answer is
"unknown".
{instruction}
{context}
{conflicts}
Question: {question}

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <your answer; if there is more than
one, separate them with |>
```

Instruction v1.

```
Some passages may contradict each other. Before
answering, check whether they do.
Ignore passages that appear to contain
misinformation, and if a disagreement reflects
genuinely different valid answers, report each one.
```

Instruction v2.

```
Some passages may contain false information. Before
answering, decide which passages are reliable by
checking them against each other.
Do not include an answer that is supported only by a
passage you judge to be unreliable.
```

Example: directional flag (passage 2 suspect).

```
Answer the question using the passages below.
If the passages support more than one valid answer
(for example, different people or things share the
same name), give every valid answer.
If no passage answers the question, the answer is
"unknown".

Passage 0 [POTENTIAL CONFLICT with passage(s): 2]:
<passage 0>

Passage 1:
<passage 1>

Passage 2 [LIKELY MISINFORMATION; conflicts with
passage(s): 0]:
<passage 2>

POTENTIAL CONFLICTS DETECTED:
- Passages 0 and 2: possible contradiction; passage
2 likely contains misinformation

Question: Who founded X?

Give a brief explanation, then end with one line in
exactly this format:
Final answer(s): <answer 1> | <answer 2> | ...
```

Directional LLM detector.

```
You will see a question and {n} numbered passages
retrieved for it.
Some passages may contain misinformation: false
claims about the answer to the question.
Identify every pair of passages that CONTRADICT each
other about the answer to the question,
i.e. they cannot both be true about the same entity
at the same time. Passages that are about
different entities with the same name, give
complementary details, or describe different times
are NOT contradictions; note them as "ambiguity",
"complementary" or "temporal" instead.

For each contradiction, say which passage of the
pair is more likely the misinformation, judging
by agreement with the other passages and by
plausibility: its number, "both" if both are likely
false, or "unclear" if you cannot tell.

Question: {question}

{passages}

Return JSON only: {{"pairs": [{{"i": <int>, "j":
<int>, "relation":
"contradiction|ambiguity|complementary|temporal",
"misinformation": <int>|"both"|"unclear"}}]}}
Use the passage numbers shown. "misinformation" is
needed only for contradictions. Return {{"pairs":
[]}} if there are none.
```

B Additional Results

Robustness to the metric and the labels. Table 5 repeats Table 1 with the bidirectional metric. Table 6 repeats the pre-registered tests with every suspect document treated as misinformation. Table 3 lists all pre-registered tests.

The heuristic detector. CAAG is the detector we set out to study. As a neutral flag it did not beat random flags of the same volume, and as a removal rule it only deleted correct passages (-7.9 points on clean questions, 0.0 on conflict ones). Its false flags fall mostly on pairs that should both be reported: 134 of its 221 false flags on the hold-out are same-name ambiguity pairs, which its entity-mismatch signal flags because they mention different dates or numbers. We read this as a warning about lexical contradiction cues in general (Gururangan et al., 2018), rather than as a problem with a particular threshold. Retuning illustrates why F1 misleads. Over a 121-setting sweep, CAAG's best pair F1 is 29.2, above NLI's, at $\theta_s = 0.40$ and

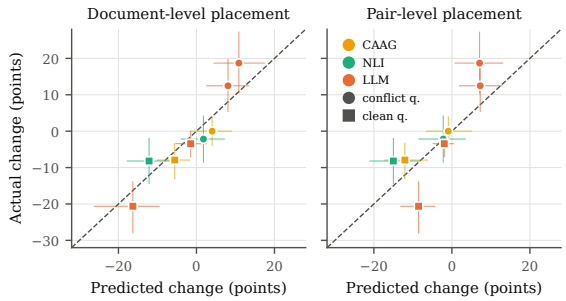


Figure 3: Effect of real detectors predicted from the synthetic grid (x, with the 95% bootstrap prediction interval) against the measured effect (y, with its 95% CI), for conflict (•) and clean (■) questions. Left: detectors located by their document-level behaviour; right: by pair precision, recall and direction accuracy.

$\theta_c = 0.10$. But it gets there by flagging 43% of all pairs and 87% of clean questions, which our grid predicts would be harmful under removal (Appendix B).

Original CAAG format. Rendering CAAG's flags in its original format, with typed lines and confidence scores ("factual conflict (confidence: 0.41)"), changes accuracy by −0.9 points overall (−1.2 on conflict questions, −1.6 on clean ones). This is no different from the uniform format.

Pilot study (exploratory). On the 200-question pilot split, scored with the bidirectional metric, the same pattern appeared before any confirmatory run. Oracle neutral flags changed conflict-question accuracy by −10.5 points, directional flags by +11.8, and removing the suspect by +28.4. The pilots also showed that the original REMOVE strategy, which deletes an arbitrary side of each pair, removes correct documents. They are why we added directional formats, synthetic direction accuracy and document-level placement. All pilot runs are released.

Phase-0 audit of CAAG on NarrativeQA. CAAG was designed for long-document QA. On 100 NarrativeQA (Kočiský et al., 2018) questions from 50 stories (BM25 top-10 chunks from a single coherent story), it flags at least one pair in 54% of questions, 166 pairs in all. A single story rarely contradicts itself, so most of these flags are likely false; the flagged pairs are released for annotation. This is the setting where a false flag can only hurt.

C Reproducibility

Runs. Every number in this paper is produced by one script from committed run outputs: per-call responses, token counts, detector outputs and configurations. The pre-registration was committed before any confirmatory run. Its deviations are listed below.

Deviations from the pre-registration.

1. *Primary metric.* The first draft of the pre-registration named the bidirectional metric as primary. Before the confirmatory run we switched to one-way matching, because it reproduces MADAM-RAG's published score on the pilot split (Section 4); bidirectional became the robustness metric. Every table reports both.
2. *Second generator.* The plan named an open-weights model. None was available before the confirmatory run, so gpt-4.1 was used (H9). After review we added Qwen2.5-7B-Instruct, registered before its run. Because local inference takes about 3 s per call, the Qwen runs use one seed; we made that change before inspecting any Qwen result.
3. *CONFLICTS judge.* The plan called for a judge from a different model family, validated on about 100 human labels. Neither the judge nor the labels were available, so CONFLICTS is scored only on items with a reference answer, with one seed, and reported as exploratory.
4. *H4.* The pre-registration counted 54 false-flag cells. On clean questions, cells that differ only in direction accuracy render identical prompts (their scores differ by exactly 0), so they are one comparison: 18 in all, 9 per format. We report H4a per format. The cell-permutation p-value for the dose-response correlation assumed exchangeable cells; the cells are nested, so we replace it with a question-level bootstrap CI. No threshold was set for H4c, so we report a count.
5. *H5.* A non-significant difference is not equivalence. We add a TOST with a ±3-point margin, fixed in the addendum.
6. *H6.* The first draft tested whether the grid prediction lies inside the actual effect's CI, which is lenient. We now report the pre-registered test (actual effect inside a bootstrap prediction interval) and a paired test of the two spaces' errors.
7. *Placement modes.* For H6, the detectors

without direction were run under the label `remove_suspect` as well as `remove`. Both render identical prompts, so these were served entirely from the response cache.

8. *Holm scope.* Table 1 applies Holm across all its comparisons, which is stricter than the per-family correction used for the hypothesis tests.

The addendum (H10, H11 and the re-analyses above) and the Qwen seed change were committed to the repository before the runs they cover. The repository's commit history is the only timestamp; we did not use a third-party registry (van Miltenburg et al., 2021).

Cost and compute. All API calls in this paper, pilots included, cost $33.70. CAAG and NLI ran on a laptop CPU (0.2 s and 1.2 s per question).

	Pre-registered test	Estimate [95% CI]	p_{Holm}	Outcome	Robust
H1	F1 llm_dir > nli	+2.6 [−2.4, +7.5]	0.311	not supported	–
H1	F1 nli > caag	+12.2 [+7.5, +16.9]	<0.001	supported	–
H1	direction accuracy llm_dir > 50% (cluster CI[r])	88.3% [83.3, 92.9]	<0.001	supported	–
H2	oracle+flag < plain (conflict)	−6.7 [−10.8, −3.1]	<0.001	supported	yes
H3	oracle+flag_directional > plain (conflict)	+13.4 [+6.7, +20.4]	<0.001	supported	yes
H3	oracle+flag_directional > oracle+flag (conflict)	+20.1 [+13.9, +26.9]	<0.001	supported	yes
H3	oracle+remove_suspect > plain (conflict)	+27.6 [+20.6, +35.0]	<0.001	supported	yes
H4a	false-flag cells below plain on clean q., directional flag[r]	0 of 9 cells	Holm	not supported	yes
H4a	false-flag cells below plain on clean q., removal[r]	6 of 9 cells	Holm	supported	yes
H4b	harm grows with clean-q. exposure, Spearman over 24 cells[r]	$\rho = -0.72$ [−0.86, −0.19]	–	supported	yes
H4c	removal harms clean q. more than directional flag[r]	5 of 9 matched cells	Holm	count only	–
H5	caag+flag vs random+flag (all)	−1.4 [−3.4, +0.3]	0.178	n.s.	yes
H5	equivalence within ±3 points (TOST)[r]	90% CI [−3.1, +0.1]	–	not equivalent	yes
H6	actual effect inside the 95% interval of the document-level prediction[r]	7 of 8 (pair level: 6 of 8)	–	descriptive	–
H6	document-level error < pair-level error (paired, exact sign-flip)[r]	−1.2 [−3.6, +0.4]	0.414	not supported	yes
H7	instruct vs plain (conflict)	−1.2 [−4.3, +1.7]	1.000	n.s.	yes
H7	instruct vs plain (clean)	+2.9 [0.0, +6.1]	0.280	n.s.	yes
H7	instruct_v2 vs plain (conflict)	+5.3 [+1.4, +9.6]	0.059	n.s.	yes
H7	instruct_v2 vs plain (clean)	−1.1 [−5.0, +2.6]	1.000	n.s.	yes
H9	gpt-4.1: oracle+flag < plain (conflict)	−8.6 [−13.2, −4.3]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > plain (conflict)	+18.2 [+12.0, +24.7]	<0.001	supported	yes
H9	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+26.9 [+20.1, +33.6]	<0.001	supported	yes
H9	gpt-4.1: oracle+remove_suspect > plain (conflict)	+30.2 [+22.5, +37.9]	<0.001	supported	yes
H8	WikiContradict: instruct vs plain	+8.7 [+5.4, +12.3]	<0.001	significant	–
H8	WikiContradict: instruct_v2 vs plain	−6.7 [−10.0, −3.7]	<0.001	significant	–
H8	WikiContradict: oracle+flag vs plain	+6.5 [+3.0, +10.0]	<0.001	supported	–
H8	WikiContradict: caag+flag vs plain	+0.7 [0.0, +1.6]	0.250	n.s.	–
H8	WikiContradict: nli+flag vs plain	+3.2 [+0.3, +6.1]	0.077	n.s.	–
H8	WikiContradict: llm_dir+flag vs plain	+6.2 [+2.8, +9.6]	<0.001	significant	–
H10	gpt-4o-mini: oracle+flag vs plain (conflict)	−5.3 [−9.4, −1.4]	0.042	significant	yes
H10	gpt-4o-mini: oracle+flag_directional > plain (conflict)	+5.0 [−1.7, +11.8]	0.314	not supported	yes
H10	gpt-4o-mini: oracle+flag_directional > oracle+flag (conflict)	+10.3 [+4.8, +16.1]	0.005	supported	yes
H10	gpt-4o-mini: oracle+remove_suspect > plain (conflict)	+18.9 [+12.5, +25.9]	0.001	supported	yes
H10	gpt-4.1: oracle+flag vs plain (conflict)	−11.0 [−17.5, −4.6]	0.010	significant	yes
H10	gpt-4.1: oracle+flag_directional > plain (conflict)	+12.7 [+5.8, +19.7]	0.005	supported	yes
H10	gpt-4.1: oracle+flag_directional > oracle+flag (conflict)	+23.7 [+17.0, +30.7]	0.001	supported	yes
H10	gpt-4.1: oracle+remove_suspect > plain (conflict)	+14.9 [+8.6, +21.6]	0.001	supported	yes
H10	Qwen2.5-7B: oracle+flag vs plain (conflict)	−5.8 [−10.8, −1.4]	0.117	n.s.	yes
H10	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	+2.9 [−3.6, +9.4]	0.509	not supported	yes
H10	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	+8.6 [+2.9, +14.4]	0.041	supported	yes
H10	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	+10.8 [+4.3, +17.3]	0.017	supported	yes
H11	Qwen2.5-7B: oracle+flag < plain (conflict)	−5.8 [−10.8, −0.7]	0.114	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > plain (conflict)	+3.6 [−0.7, +8.6]	0.224	not supported	yes
H11	Qwen2.5-7B: oracle+flag_directional > oracle+flag (conflict)	+9.4 [+5.0, +14.4]	0.001	supported	yes
H11	Qwen2.5-7B: oracle+remove_suspect > plain (conflict)	+18.0 [+10.8, +25.2]	<0.001	supported	yes

Table 3: Every pre-registered test on the confirmatory split (one-way strict accuracy; differences in points). Holm correction within each family. H1–H9 were registered before the confirmatory run; H10, H11 and the rows marked [r] were registered after a review of the first draft (H10, H11 before their runs; [r] rows are re-analyses of data that already existed). Directional predictions are *supported* or *not supported*; tests with no predicted sign are *significant* or *n.s. Robust*: the same outcome under the bidirectional metric.

	Pair level			Clean-q.	Direc-	Removal, per question		Latency	Cost
Detector	P	R	F1	flagged	tion	misinfo gone	correct lost	(s)	($/1k q.)
CAAG (heuristic)	21.7	7.9	11.6	34.1	–	9.4	31.7	0.23	0.000
NLI cross-encoder	20.1	29.3	23.8	61.9	–	44.6	52.4	1.18	0.000
LLM (gpt-4o-mini)	37.2	20.5	26.4	76.2	88.3	61.2	65.9	0.80	0.181

Table 4: Detectors on the RAMDocs hold-out split (300 questions; percentages). *Clean-q. flagged*: share of conflict-free questions with at least one flag. *Direction*: share of correctly flagged contradictions whose suspect side is the misinformation. *Misinfo gone*: conflict questions whose misinformation documents are all removed when removing the suspect (or, without direction, the second passage). *Correct lost*: clean questions that lose at least one correct document. Latency is per question; CAAG and NLI run locally.

Condition	All	Lab.	Conflict	Clean	Tokens
No flags (plain)	35.2	37.2	25.9	49.7	911
Prompt only					
Instruction (v1)	+1.3	+0.5	−1.0	+2.1	948
Instruction (v2)	+2.2	+1.4	+3.6	−1.1	951
Oracle detector					
Oracle, flag	**−3.6**	**−4.0**	**−7.7**	0.0	980
Oracle, directional flag	**+7.0**	**+7.9**	**+15.1**	0.0	998
Oracle, remove suspect	**+15.4**	**+17.5**	**+33.3**	0.0	825
Real detectors (exploratory)					
Random, flag	+0.4	+0.4	−0.2	+1.1	949
CAAG, flag	−0.9	−1.4	−1.9	−0.8	942
CAAG, remove	−4.0	−4.5	−1.0	−8.5	809
NLI, flag	+0.2	0.0	−1.4	+1.6	1021
NLI, remove	−8.0	−7.9	−3.1	**−13.2**	592
LLM, flag	−3.4	−3.4	**−7.2**	+0.8	970
LLM, directional flag	+4.7	+4.5	**+14.1**	−6.1	983
LLM, remove suspect	−2.1	−2.6	**+18.9**	**−26.5**	718
Reference					
MADAM-RAG	+5.7	+5.2	+3.6	+6.9	8196

Table 5: RAMDocs hold-out, gpt-4o-mini, 3 seeds, bidirectional strict accuracy (robustness). *All*: 300 questions, including 35 whose conflict status the label rule leaves open; *Lab.*: the 265 labelled ones (139 conflict, 126 clean). First row: accuracy (%) with no flags. Other rows: paired difference from it in points; **bold** = significant after Holm correction across the table (sign-flip test, $p < 0.05$). Real-detector rows were not pre-registered as tests. Tokens: mean prompt tokens per call (MADAM-RAG: per question, all calls).

	Test (labels: *contains*)	One-way [95% CI]	p_{Holm}	Bidir.	p_{Holm}
H2	oracle+flag < plain (conflict)	−4.6 [−8.1, −1.5]	0.006	−5.2	0.003
H3	oracle+flag_directional > plain (conflict)	+11.6 [+5.6, +17.3]	<0.001	+13.3	<0.001
H3	oracle+flag_directional > oracle+flag (conflict)	+16.2 [+10.8, +21.8]	<0.001	+18.5	<0.001
H3	oracle+remove_suspect > plain (conflict)	+25.8 [+19.5, +32.2]	<0.001	+30.3	<0.001
H4	harmful cells, clean q. (dir. flag; removal); ρ	0/9; 8/9; $\rho = -0.51$ [-0.79, -0.08]			

Table 6: Label robustness: pre-registered tests with every suspect document treated as misinformation (173 conflict and 127 clean hold-out questions).

CAAG setting	θ_s	θ_c	P	R	F1	Clean FPR
Default	0.65	0.35	21.7	7.9	11.6	34.1
Best pair F1	0.40	0.10	20.7	49.6	29.2	87.3
Best question F1	0.55	0.15	22.2	28.8	25.1	60.3

Signals (defaults)	Pair F1	Flag rate	Clean FPR
all signals	11.6	6.2	34.1
negation only	0.6	0.4	4.0
entity only	8.6	4.3	23.0
temporal only	0.0	0.0	0.0

Table 7: CAAG on the RAMDocs hold-out (%). Top: thresholds from a sweep over 121 settings ($\theta_s \in [0.40, 0.90]$, $\theta_c \in [0.10, 0.60]$). Bottom: each signal alone.

Condition	All	Conflict	No conflict
No flags (plain)	66.7	49.3	73.8
Instruction (v2)	−0.4	+1.4	−1.2
CAAG, flag	+0.4	+1.4	0.0
NLI, flag	−1.7	−4.3	−0.6
LLM, flag	+1.7	+2.9	+1.2
LLM, directional flag	−0.4	−2.9	+0.6

Table 8: CONFLICTS, 237 automatically scorable questions (69 conflict), gpt-4o-mini, one seed; exploratory. Contains-match accuracy (%) and differences in points; no difference is significant (bold would mark raw $p < 0.05$). CAAG: flags 94.2% of conflict and 92.9% of clean questions; NLI: flags 52.2% of conflict and 43.5% of clean questions; LLM: flags 98.6% of conflict and 83.9% of clean questions.

Condition	All	Lab.	Conflict	Clean
No flags (plain)	32.7	21.8	48.4	
Instruction (v2)	**+5.8**	**+12.2**	−2.6	
Oracle, flag	**−4.0**	**−8.6**	0.0	
Oracle, directional flag	**+8.4**	**+18.2**	0.0	
Oracle, remove suspect	**+14.0**	**+30.2**	0.0	
LLM, directional flag	**+5.4**	**+8.4**	0.0	
LLM, remove suspect	+1.0	**+20.6**	**−24.3**	
NLI, remove	**−7.0**	+0.2	**−15.6**	

Table 9: Replication with gpt-4.1 as generator (RAMDocs hold-out, 3 seeds, one-way strict accuracy; detector fixed to gpt-4o-mini). Same conventions as Table 1.

Condition	Both answers
No flags (plain)	14.0
Instruction (v1)	**+8.7**
Instruction (v2)	**−6.7**
Oracle, flag	**+6.5**
CAAG, flag	+0.7
NLI, flag	+3.2
LLM, flag	**+6.2**

Table 10: WikiContradict (253 genuine conflicts between two Wikipedia passages; the reference answer is both sides), gpt-4o-mini, 3 seeds. First row: share of answers giving both sides (%); other rows: difference in points, **bold** = significant after Holm within H8.